

Total No. of Questions : 8]

SEAT No. :

PE-2172

[Total No. of Pages : 2

[6584]-71

B.E. (Civil)

**INTEGRATED WATER RESOURCES PLANNING AND
MANAGEMENT**

(2019 Pattern) (Semester - VII) (401003 C) (Elective-III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain causes prevention & control of surface & ground water pollution. **[10]**

b) Explain the concept of polluters pay principle along with an example. **[8]**

OR

Q2) a) What is Environmental Impact Assessment (EIA) and what are its Objectives, state methodology to carryout EIA. **[10]**

b) State any four regulations of Central Pollution Control Board (CPCB) related to water pollution and explain need of its strict implementation and monitoring. **[8]**

Q3) a) Why water is considered as an economic good? State and explain economic value of water. **[10]**

b) State and explain principles of planning and financing of water resources projects. **[7]**

OR

Q4) a) What are the causes and effects of water scarcity and control measures for preventing water scarcity in India? **[10]**

b) Explain in brief economics and decision making in IWRPM. **[7]**

P.T.O.

Q5) a) State and explain UN laws on non-navigable uses of international water courses. **[10]**

b) State and explain Dublin Principles (1992) in detail. **[8]**

OR

Q6) a) What are water rights and state the water allocation priorities in detail. **[10]**

b) State any four municipal corporation laws regarding water supply and drainage. **[8]**

Q7) a) Explain in detail flood forecasting? What are the methods to forecast flood? **[10]**

b) What are the effect of hydraulic structures on river surface profiles and sediment transport? **[7]**

OR

Q8) a) Explain in detail role of dam in flood control and power generation and its importance in IWRPM. **[10]**

b) Explain application of soft computing in flood control. **[7]**

